

Exercise on "Design of a PVC gutter"

A rain gutter (in Danish: tagrende) made of PVC is 12 m long.
The heat expansion coefficient of the PVC is $\alpha = 70 \times 10^{-6} \text{ m}/(\text{m} \cdot ^\circ\text{C})$.

1. Calculate the thermal expansion of the gutter for a temperature increase from 20 °C to 60 °C in millimetres.
2. Calculate the strain in case the thermal expansion was prevented.
3. Compare this strain to the maximal usable strain for the material.
4. Calculate the necessary force to prevent the thermal expansion just after the temperature reached 60 °C if the relevant short term modulus is 1500 MPa and the cross section of the gutter has an area of 400 mm².
5. Calculate the necessary force to prevent the thermal expansion 1 h after the temperature reached 60 °C and 100 h after the temperature reached 60°C.



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The heat expansion coefficient of the pvc is $\alpha = 70 \times 10^{-6} \text{ meter}/(\text{meter degree C})$.

- a. Calculate the thermal expansion of the gutter be for a temperature increase from 20 to 60 degrees C be, measured in millimetres?
- b. Calculate the strain if the thermal expansion was prevented?
- c. Compare this strain to maximal usable strain for the material?
- d. Calculate the necessary force to prevent the thermal expansion if the relevant modulus is 1500 MPa and the cross section of the gutter has an area of 400 mm²?

$$\Delta l = l \times \alpha \times \Delta T = 12000 \times 70 \times 10^{-6} (60-20) = 33,6 \text{ mm.}$$

$$\epsilon = \Delta l / l = 33,6 / 12000 = 0,0028 = 0,28 \%$$

$$\epsilon_{\text{max}} = 0,8 \%$$

$$F = E \epsilon A = 1500 \times 0,0028 \times 400 \text{ N} = 1680 \text{ N}$$

$$E_c(\text{PVC}, 1\text{h}) = 1000 \text{ MPa} \rightarrow F(1\text{h}) = 1000 \times 0,0028 \times 400 = 1120 \text{ N}$$

$$E_c(\text{PVC}, 100\text{h}) = 560 \text{ MPa} \rightarrow F(100\text{h}) = 560 \times 0,0028 \times 400 = 627 \text{ N}$$

Note: data of PVC material from Table 4 and Table 8 in 'How to select the right plastic material' (2006) by E. M. Kjær.